

Mohan Krishna Srinivasula

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Mathematics

Expected Graduation: May 2028

Relevant Coursework: Database Systems, Data Structures & Algorithms, Statistical Analysis, Computer Systems

Leadership: UIUC Campus Ambassador @ InternDB, SIG AI Project Lead @ UIC Association Computing Machinery

EXPERIENCE

Undergraduate Research Assistant

June 2026 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Onboarding onto **ScribeAR**, a real-time captioning system for multi-platform (augmented reality inclusive) accessibility under Professor Lawrence Angrave, tracing its audio pipeline from **WebSocket** ingestion through a multiprocess **faster-whisper** transcription service to production deployment.

Software Engineering Intern

June 2025 – Aug. 2025

University of Illinois Chicago (MaSE Lab)

Chicago, IL

- Built a closed-loop **microscope stage automation system** in Python, integrating **ToupCam SDK** and **Elveflow** hardware APIs for autonomous mixer manifold traversal at micron-level precision.
- Developed a real-time **computer vision pipeline** achieving **95%+** mixer hole-detection accuracy via **OpenCV** contour analysis and geometric center-finding, cutting processing time by **86% (14s → 2s)**.
- Implemented a multithreaded snake-scan **traversal algorithm** with edge-case handling for flat regions and manifold boundaries, reducing manual alignment time by **70%** across **100+** experimental samples.

Freelance Software Engineer

Mar. 2024 – Aug. 2024

AiC Company

Remote

- Built a pay-per-view gated checkout through **Stripe API integration** for an HLS video player, gating **10+** language-specific streams behind purchase through Node.js server-side payment intents and webhook handlers.
- Built a **7-page** responsive e-commerce store in **JavaScript/HTML/CSS**, reducing user drop-off by **20%**.
- Developed a user registration flow with **Twilio SMS 2FA** on a Node.js backend, deployed through **AWS**.

PROJECTS

Prism (HackIllinois 2026) | Python, TypeScript, RAG, Claude, Pinecone, GithubAPI, FastAPI | [Link](#)

- Built a **5-stage** AI agentic pipeline autonomously converting **1-2 star** app reviews into GitHub PRs to fix issues.
- Built a RAG retrieval system for the **top 20** relevant functions with Voyage AI and **Pinecone vector database**.
- Tested Claude-generated patches** in isolated Modal Sandboxes (gVisor) through a Ruff/ktlint and **Pytest self-correction loop**, only opening PRs after all checks pass.

Multithreaded HTTP Server | C, Linux, Sockets, POSIX, ApacheBench | [Link](#)

- Implemented an **HTTP/1.1 server from scratch in C** using POSIX sockets (getaddrinfo, bind, listen, accept), handling GET requests and serving HTML static files with proper header construction.
- Built request parsing and file-serving logic with dynamic content-length calculation and **404 error** handling.
- Sustained **8,833 req/sec** at **200 concurrent connections** with **zero failed requests** through a thread-per-connection model with pthreads (benchmarked via ApacheBench, **10,000 requests**).

Geospatial Photo-Sharing Platform | Python, SQL, MySQL, Flask, Docker, React

- Designed a normalized (**3NF**) **MySQL schema** across **8 tables**, applying constraints, **ingesting data** through a Python script, and **modeling data** from the Flickr Geotagged (~**10K**) and GeoNames (~**2.3M**) datasets.
- Built a Flask backend with REST endpoints, serving bounding-box spatial queries over **2,000+** geotagged photos.
- Implemented a trigger prior to photo insertion normalizing **1000+** photo geocoordinates into valid lat/long ranges and repeatable read transactions wrapping **multi-query dashboard** aggregations for map snapshot consistency.
- Reduced advanced query cost by up to **99% (60.9 → 0.35)** and **58% (165 → 69)** through indexing plans.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C, C++, JavaScript, R, HTML/CSS

Frameworks & Backend: React, Node.js, Next.js, Flask, FastAPI, REST APIs, PySpark

Data & Cloud: Azure, MySQL, PostgreSQL, AWS, Tableau, Data Pipelines, Data Modeling

Tools & AI: Linux, Git, Docker, CI/CD, LangChain, RAG, LLMs, NLP, ApacheBench